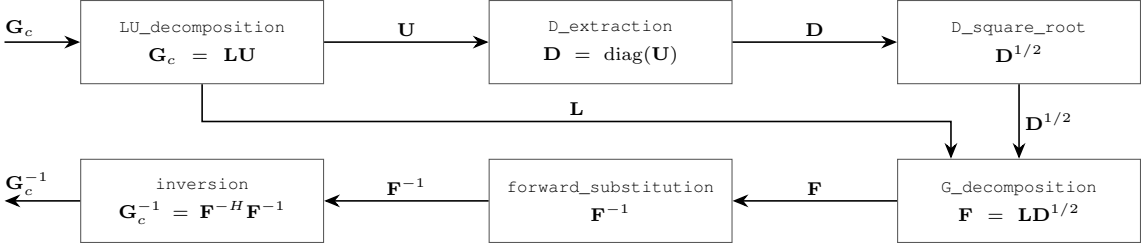
$\mathbf{G}_c$
LU_decomposition
$\mathbf{G}_c = \mathbf{LU}$
$\mathbf{U}$
D_extraction
$\mathbf{D} = \mathrm{diag}(\mathbf{U})$
$\mathbf{D}$
D_square_root
$\mathbf{D}^{1/2}$
$\mathbf{L}$
$\mathbf{D}^{1/2}$
$\mathbf{G}_c^{-1}$
inversion
$\mathbf{G}_c^{-1} = \mathbf{F}^{-H}\mathbf{F}^{-1}$
$\mathbf{F}^{-1}$
forward_substitution
$\mathbf{F}^{-1}$
$\mathbf{F}$
G_decomposition
$\mathbf{F} = \mathbf{L}\mathbf{D}^{1/2}$